

Definition: Random Definitions

- $\text{pr}_i: X_1 \times \dots \times X_n \rightarrow X_i, (x_1, \dots, x_n) \mapsto x_i$
- The **power set** of X is $\mathcal{P}(X) = \{U: U \subseteq X\}$
- A subset of $\mathcal{P}$ is a **system of subsets** of X .
- $\emptyset$ is understood to be the linear combination of no vectors.
- A (an operator/mat/etc) is **nilpotent** w degree k if $A^k = 0, k \in \mathbb{N}$.
- For a set X and $f: X \rightarrow X, Y \subseteq X$ is **stable** under f if $f(Y) \subseteq Y$.
- $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n, n \in \mathbb{N}$ is a **linear combination** of $\vec{v}_1, \dots, \vec{v}_n$.

Theorem: Random Theorems

- Group axioms for $(G, *)$: G closed under $*$; $*$ associative $[g*(h*k) = (g*h)*k]$, $\exists$ unique identity $\text{id} \in G$; $\exists$ unique inverse $g^{-1} \in G \forall g$.
- For G, H grps, $f: G \rightarrow H$ is a group hom. if $f(ab) = (f(a)f(b))$.
- For f a group hom. $\ker(f) = \{a \in G: f(a) = e\}$ is a normal subgroup of G .
- O-S: For G a group acting on a finite set $X, |\text{Orb}(x)| = |G|/|\text{Stab}(x)|$
- Lagrange's Theorem: For G a finite group and $H \leq G$ a normal subgroup (invariant under conjugation, $ghg^{-1} \in H \forall g \in G, h \in H$); $|G| = |H| |G/H|$

Remark: Notation

- VS = Vector Space. VsubS = Vector subspace. $F-VS = F$ - Vector Space
- 'lin-ind' = 'linearly independent' and 'lin-dep' = 'linearly dependent'
- 'almost all' means 'all but finitely many'
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Theorem: Properties Of Ideals

- $\ker f$ is an ideal of R .
- The intersection of any collection of ideals is an ideal
- For I, J ideals of R ; $I + J := \{a+b : a \in I, b \in J\}$ is an ideal
- For R a comring and $T \subseteq R$, the ideal I of R generated by T is the set $R(T) = \{r_1 t_1 + \dots + r_m t_m : t_1, \dots, t_m \in T, r_1, \dots, r_m \in R\}$ together with 0_R if $T = \emptyset$.
- If T is finite can write $R(t_1, \dots, t_n)$.
- For $m \in \mathbb{Z}$, $\mathbb{Z}(m) = m\mathbb{Z}$
- For R a comring and $T \subseteq R$, $R(T)$ is the smallest ideal of R containing T .
- For R a comring, an ideal I is a **principal ideal** if $I = (t)$ for some $t \in R$.
- In a comring, $\{0\} = (0)$ and $R = (1_R)$ are always principal ideals.

Definition: Equivalence and Equivalence Relations

A relation R on a set X is a subset $R \subseteq X \times X$.
Only in this context, $(x, y) \in R$ is written xRy .
 R is an **equivalence relation** on X when for all $x, y, z \in X$:

- Reflexivity: xRx
- Symmetry: xRy if and only if yRx
- Transitivity: $(xRy \text{ and } yRz) \implies xRz$

* For $\sim$ an equivalence relation on X and $x \in X$; the set $E(x) := \{z \in X : z \sim x\}$ is the **equivalence class** of x .

- $E \subseteq X$ is an **equivalence class** for $\sim$ if $\exists x \in X$ s.t. $E = E(x)$.
- An element of an equivalence class is a **representative** of the class.
- $Z \subseteq X$ is a **system of representatives** for $\sim$ if it contains precisely one element from each equivalence class of $\sim$.
- For $x, y \in X$ TFAE: $x \sim y$; $E(x) = E(y)$; $E(x) \cap E(y) \neq \emptyset$
- The set of equiv classes of X is $(X/\sim) := \{E(x) : x \in X\} \subseteq \mathcal{P}(X)$.**
- There is a surjective canonical mapping $\text{can} : X \rightarrow (X/d), x \mapsto E(x)$.

Definition: Quotient Vector Space

For V an F -VS, $W \subseteq V$ a VsubS, $\sim$ on V s.t. $x \sim y$ if $x - y \in W$, so for $x \in V$, $E(x) = W + x \subseteq V$. The **quotient VS** of V by W , $V/\sim = V/W$ is an F -VS with $\lambda E(x) = E(\lambda x)$ and $\text{can} : V \rightarrow V/W$ is surj. with $\ker = W$. If V finite-dim, so are W and V/W , and $\dim(V/W) = \dim(V) - \dim(W)$.

Thm: Universal Prop. Of The Set Of Equivalence Classes

Let $\sim$ be an equivrel on X .

If $f : X \rightarrow Z$ is a mapping s.t. $x \sim y \implies f(x) = f(y)$; there is a unique mapping $\bar{f} : (X/\sim) \rightarrow Z$, $\bar{f}(E(x)) := f(x)$, and $f = \bar{f} \circ \text{can}$.
Often when defining a map $g : (X/\sim) \rightarrow Z$, we define $f : X \rightarrow Z$ and say g is **well-defined** to indicate that $x \sim y \implies f(x) = f(y)$ as this guarantees the existence of $g = \bar{f}$.

Definition: Cosets and Factor Rings

Let R be a ring, $I \trianglelefteq R$ an ideal of R , and $\sim$ defined by $x \sim y \iff x - y \in I$.
 $x + I := \{x + i : i \in I\} \subseteq R$
is a **coset** of I in R or the **coset** of x w.r.t. I in R
The set R/I , the **factor ring** of R by I or **quotient** of R by I , is the set $(R/\sim)$ of cosets of I in R .

- R/I is a ring, with addition defined by $(x+I) + (y+I) = (x+y)+I$ and multiplication by $(x+I)(y+I) = xy+I$ for all $x, y \in R$.
- If R is commutative so is R/I
- R/I is a nonzero ring if and only if $I \neq R$
- If I is proper, and $r \in R^\times$; $r+I \in (R/I)^\times$ with $(r+I)^{-1} = r^{-1}+I$.

Theorem: Universal Property Of Factor Rings

Let R, S be rings and $I \trianglelefteq R$. Then:

- The mapping $\text{can} : R \rightarrow R/I, r \mapsto r+I$ is a surjective ring hom. with kernel I .
- For $f : R \rightarrow S$ a ring hom. with $f(I) = \{0_S\}$, ie $I \subseteq \ker f$, there exists a unique ring hom. $\bar{f} : R/I \rightarrow S$ s.t. $f = \bar{f} \circ \text{can}$.

Major Theorem: The First Isomorphism Theorem For Rings

Let R, S be rings. Then every ring hom. $f : R \rightarrow S$ induces a ring iso.: $\bar{f} : R/\ker f \xrightarrow{\sim} \text{im } f$

Definition: Factor Modules

Let R be a ring, M an R -module, and N a submodule of M .
For $a \in M$, the **coset** of a w.r.t. N in M is : $a+N = \{a+b : b \in N\}$.

- It is a coset of N in the abelian group M , so is an equiv class for the equivrel $a \sim b \iff a - b \in N$.
- M/N , the **factor/quotient** of M by N is the set $(M/\sim)$ of all cosets of N in M .
- This becomes an R -module by defining: $(a+N) + (b+N) := (a+b)+N$ and $r(a+N) = ra+N$, for all $a, b \in M, r \in R$.
- The R -module M/N is the **factor module** of M by the submodule N .

Theorem: Universal Property Of Factor Modules

Let R be a ring, L, M be R -modules, and N a submodule of M :

- The mapping $\text{can} : M \rightarrow M/N, a \mapsto a+N$ for all $a \in M$ is a surjective R -hom. with kernel N .
- If $f : M \rightarrow L$ is an R -hom with $f(N) = \{0_L\}$ so that $N \subseteq \ker f$, there exists unique hom. $\bar{f} : M/N \rightarrow L$ such that $f = \bar{f} \circ \text{can}$.

Major Theorem: First Isomorphism Theorem For Modules

Let R be a ring and M, N be R -modules.
Then every R -hom. $f : M \rightarrow N$ induces an R -iso. $\bar{f} : M/\ker f \xrightarrow{\sim} \text{im } f$

- 2nd** Theorem: For K, L sub-mods of M ; L is a submod of $K+L = \{b+c : b \in K, c \in L\}$ and $K \cap L$ is a submod of K . Also, $\frac{K+L}{K \cap L} \cong \frac{K}{K \cap L} \oplus \frac{L}{K \cap L}$
- 3rd** Theorem: Let K, L be sub-mods of M , with $L \subseteq K$. Then K/L is a submod of M/L and $\frac{M/L}{K/L} \cong M/K$.

Determinants And Eigenvalues

Definition: Permutations And Their Signs

The group (under composition) of all permutations of the set $\{1, \dots, n\}$ to itself, denoted $\mathfrak{S}_n$, is the n -**th symmetric group**. It has order $|\mathfrak{S}_n| = n!$.

- A **transposition** is a perm that swaps two elements (no effect on others)
- An **inversion** of a $\sigma \in \mathfrak{S}_n$ is a pair (i, j) s.t. $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$.
- The number of inversions of a σ is the **length** of σ , $\ell(\sigma)$
- The **sign** of σ is the parity of the length of σ : $\text{sgn}(\sigma) = (-1)^{\ell(\sigma)}$
- If $\text{sgn}(\sigma) = 1/-1$, σ is an **even/odd permutation**
- $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$
- The set of even perms in $\mathfrak{S}_n$ forms a subgroup (it is the kernel of $\text{sgn}(\sigma)$, a grp hom.) of $\mathfrak{S}_n$ called the **alternating group** and denoted A_n .

Definition: Cofactors And Adjugate Matrices

Let $A \in \text{Mat}(n; R)$ for a comring R , $n, i, j \in \mathbb{N}$ s.t. $1 \leq i, j \leq n$. Then:

- The (i, j) **cofactor** of A is $C_{ij} = (-1)^{i+j} \det(A(i, j))$ where $A(i, j)$ is the matrix obtained by deleting the i^{th} row and j^{th} column of A .
- The **adjugate matrix** $\text{adj}(A)$ is the $(n \times n)$ matrix whose entries are $\text{adj}(A)_{ij} = C_{ji}$ (note order swap of i and j !).

Major Definition: The Determinant

Let R be a comring and $n \in \mathbb{N}$. The **determinant** is a mapping $\det : \text{Mat}(n; R) \rightarrow R$ given by: $\det(A) = \sum_{\sigma \in \mathfrak{S}_n} \text{sgn}(\sigma) a_{1\sigma(1)} \dots a_{n\sigma(n)}$. This is called the **Leibniz formula**. For $n=0$ $\det(A) = 1$.

Alternative characterisations are:

- Let F be a field. The mapping $\det : \text{Mat}(n; F) \rightarrow F$ is the unique alternating multilinear form on n -tuples of column vectors with values in F that takes the value 1_F on the identity matrix.
- Laplace Expansions: Let R be a comring and $A = (a_{ij}) \in \text{Mat}(n; R)$. For fixed i, j the i^{th} row/ j^{th} column expansion of the determinant is $\det(A) = \sum_{j=1}^n a_{ij} C_{ij} / \sum_{i=1}^n a_{ij} C_{ij}$.

Definition: Bi/Multilinear Forms

Let U, V, W be F -VSs. A **bilinear form** on $U \times V$ with values in W is a mapping $H : U \times V \rightarrow W$ linear in both entries. Ie, $\forall u_1, u_2 \in U, v_1, v_2 \in V$:

- $H(\lambda u_1, v_1) = \lambda H(u_1, v_1)$
- $H(u_1 + u_2, v_1) = H(u_1, v_1) + H(u_2, v_1)$
- $H(u_1, \lambda v_1) = \lambda H(u_1, v_1)$
- $H(u_1, v_1 + v_2) = H(u_1, v_1) + H(u_1, v_2)$

For $U=V$, a bilinear form is **symmetric** if for all $u, v \in U$, $H(u, v) = H(v, u)$, and it is **alternating/antisymmetric** if for all $u \in U$, $H(u, u) = 0$

- If H is antisymmetric, $H(u, v) = -H(v, u)$
- Multilinear forms** are just an extension of bilinear forms to higher dimensions - H must be linear in each argument.
- A multilinear form is **alternating** if $\exists i \neq j$ with $v_i = v_j \implies H(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = 0$ and if so, has the property $H(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = -H(v_1, \dots, v_j, \dots, v_i, \dots, v_n)$ for all $v_1, \dots, v_n \in V$.
- Thus for any $\sigma \in \mathfrak{S}_n$, $H(v_{\sigma(1)}, \dots, v_{\sigma(n)}) = \text{sgn}(\sigma) H(v_1, \dots, v_n)$

Theorem: Cramer's Rule and Jacobi

Cramer: Let R be a comring, $A \in \text{Mat}(n; R)$. Then $A \cdot \text{adj } A = (\det(A)) I_n$
Jacobi: For $A = (a_{ij})$, $a_{ij} = a_{ij}(t)$ func of t ; $\frac{d}{dt} \det(A) = \text{tr } \text{adj } A \frac{dA}{dt}$

Major Definition: Eigenvalues and Eigenvectors

Let $f : V \rightarrow V$ be an endo. of an F -VS V .
 $\lambda \in F$ is an **eigenvalue** of f iff there exists non-z $\vec{v} \in V$ s.t. $f(\vec{v}) = \lambda \vec{v}$.
Each such $\vec{v} \in V$ is called an **eigenvector** of f with **eigenvalue** λ .
 $E(\lambda, f)$, the **eigspace** of f with **eigval** λ , is $\{\vec{v} \in V : f(\vec{v}) = \lambda \vec{v}\}$.

- Each endo. of a n -z finite-dim VS over an algly-cld field has an eigval
- For f with distinct eigenvals $\lambda_1, \dots, \lambda_n$, the eigvecs $\vec{v}_1, \dots, \vec{v}_n$ are lin-ind.

Definition: The Characteristic Polynomial

Let R be a comring and $A \in \text{Mat}(n; R)$. The polynomial $\chi_A(x) := \det(xI_n - A) \in R[x]$ is called the **characteristic polynomial** of A .

- For F a field, $A \in \text{Mat}(n; F)$; the eigvals of A are the roots of χ_A .

Theorem: Cayley-Hamilton Theorem

Let R be a comring and $A \in \text{Mat}(n; R)$.
Then $\chi_A(x)$ evaluated at the matrix A , $\chi_A(A)$, equals zero.

Definition: Conjugacy

For R a comring, $A, B \in \text{Mat}(n; R)$:

- A and B are **conjugate** if $B = P^{-1}AP$ for invertible $P \in \text{GL}(n; R)$.
- Conjugacy is an equivalence relation on $\text{Mat}(n; R)$
- The characteristic polynoms of conjugate matrices are the same

Definition: Triangularisability

Let $f : V \rightarrow V$ be an endo. of a finite-dim F -VS V . The following two statements are equivalent, and if they hold, f is **triangularisable**:

- $\chi_f(x)$, decomposes into linear factors in $F[x]$
- V has an ordered basis $\mathcal{B} = (\vec{v}_1, \dots, \vec{v}_n)$ s.t. $f(\vec{v}_1) = a_{11}\vec{v}_1, f(\vec{v}_2) = a_{12}\vec{v}_1 + a_{22}\vec{v}_2, \dots, f(\vec{v}_n) = a_{1n}\vec{v}_1 + \dots + a_{nn}\vec{v}_n \in V$. Equivalently, the matrix $[f]_{\mathcal{B}}$ is upper triangular.

- An endo. $A : F^n \rightarrow F^n$ is triangable iff A is conj to an upper triang mat B . Ie, any endo. of a finite-dim VS over an algly-cld field is triangable.

Definition: Diagonalisability

- An endo. $f : V \rightarrow V$ of an F -VS is **diagonalisable** iff there exists a basis of V made up of eigvecs of f .
- If V is finite-dim, this is equivalent to: $\exists$ ordered basis $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_n\}$ such that $[f]_{\mathcal{B}} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Here $f(\vec{v}_i) = \lambda_i \vec{v}_i$.
- A square matrix $A \in \text{Mat}(n; F)$ is **diagonalisable** iff the linear mapping $F^n \rightarrow F^n$ given by left multiplication by A is diagonalisable.
- This is, A is conjugate to a diagonal matrix ($\exists P \in \text{GL}(n; F)$ s.t. $P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n)$). In this case the columns of P are the vecs of a basis of F^n consisting of eigenvecs of A with eigenvals $\lambda_1, \dots, \lambda_n$.

Inner Product Spaces

Major Definition: Inner Products And Their Spaces

Let V be an $\mathbb{R}$ -VS. An **inner product** on V is a mapping $(-, -) : V \times V \rightarrow \mathbb{R}$ that satisfies, for all $\vec{x}, \vec{y}, \vec{z} \in V$ and $\lambda, \mu \in \mathbb{R}$:

- $(\lambda \vec{x} + \mu \vec{y}, \vec{z}) = \lambda(\vec{x}, \vec{z}) + \mu(\vec{y}, \vec{z})$
- $(\vec{x}, \vec{y}) = (\vec{y}, \vec{x})$ (together with (1) implies symmetric bilinear form)
- $(\vec{x}, \vec{x}) \geq 0$, $= 0$ iff $\vec{x} = 0$ (positive definite)
- The **standard inner product** on $V = \mathbb{R}^n$ is the dot product
- A **real inner product space** is a real VS endowed with an inner product
- For V a VS over $\mathbb{C}$ the definition is similar, swap $\mathbb{R}$ for $\mathbb{C}$ everywhere; condition (2) reads $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$; and the standard inner product is $(\vec{v}, \vec{w}) = v_1 \bar{w}_1 + \dots + v_n \bar{w}_n$. Note the SIP is not $\mathbb{C}$ -linear in the second variable.
- In a $\mathbb{R}/\mathbb{C}$ -VS, the **length/inner product norm/norm** is $\|\vec{v}\| := \sqrt{(\vec{v}, \vec{v})}$
- Vectors of length 1 are **units**
- $\vec{v}, \vec{w}$ are **orthogonal**, $\vec{v} \perp \vec{w}$, iff $(\vec{v}, \vec{w}) = 0$.
- Sets S, T are orthogonal, $S \perp T$, if $\vec{v} \perp \vec{w} \forall \vec{v} \in S, \vec{w} \in T$.
- An **orthonormal family** of vecs in an IPS is a $(\vec{v}_i)_{i \in I}$ s.t. all $\vec{v}_i$ have $\|\vec{v}_i\| = 1$ and are pairwise orthogonal (by Kronecker delta, $(\vec{v}_i, \vec{v}_j) = \delta_{ij}$)
- An orthonormal family that is a basis is an **orthonormal basis**. For V an IPS, $(\vec{v}_i)_{i \in I}$ an orthonormal basis; $\forall \vec{w} \in V$, $\vec{w} = \sum_{i \in I} (\vec{w}, \vec{v}_i) \vec{v}_i$.
- Every finite-dim IPS has an orthonormal basis

Dfn: Orthogonal Complements/Projections

Let V be an IPS, $T \subseteq V$, and U be a finite-dim subspace of V .
 $T^\perp = \{\vec{v} \in V : \vec{v} \perp \forall \vec{t} \in T\}$ is the **orthogonal** to T .
 $T^\perp$ is always a subspace of V even if T is not, and $T^\perp = \langle T \rangle^\perp$.
The space $U^\perp$ is the **orthogonal complement** to U .

- Every $\vec{v} \in V$ can be written as $\vec{v} = \vec{p} + \vec{r}$, $\vec{p} \in U$ and $\vec{r} \in U^\perp$.
- U and $U^\perp$ are complementary, ie $V = U \oplus U^\perp$.
- The **orthogonal projection from V onto U** is $\pi_U : V \rightarrow V, \vec{v} = \vec{p} + \vec{r} \mapsto \vec{p}$.
- π_U is a linmap with $\text{im}(\pi_U) = U$ and $\ker(\pi_U) = U^\perp$
- $\pi_U^2 = \pi_U$, ie π_U is idempotent
- Let $\{\vec{v}_1, \dots, \vec{v}_n\}$ be an orthonormal basis of U .
- Then for all $\vec{v} \in V$, $\pi_U(\vec{v}) = \sum_{i=1}^n (\vec{v}, \vec{v}_i) \vec{v}_i$.

Major Theorem: The Cauchy-Schwarz Inequality

Let $\vec{v}, \vec{w}$ be vectors in an inner product space.
Then $|\langle \vec{v}, \vec{w} \rangle| \leq \|\vec{v}\| \|\vec{w}\|$ with equality iff $\vec{v}, \vec{w}$ are linearly dependent.

For V an IPS, $\vec{v}, \vec{w} \in V$, and λ a scalar (ie in the underpinning field):

- $\|\vec{v}\| \geq 0$, $= 0$ iff $\vec{v} = \vec{0}$.
- $\|\lambda \vec{v}\| = |\lambda| \|\vec{v}\|$
- $\|\vec{v} + \vec{w}\| \leq \|\vec{v}\| + \|\vec{w}\|$.
- In a real IPS, $\exists$ unique $0 \leq \theta \leq \pi$ s.t. $\cos(\theta) = \frac{(\vec{v}, \vec{w})}{\|\vec{v}\| \|\vec{w}\|}$.

Theorem: Gram-Schmidt Process

Let $\vec{v}_1, \dots, \vec{v}_k$ be linearly ind. vectors in an IPS V . Then we can construct an orthonormal family $\vec{w}_1, \dots, \vec{w}_k$ such that for all $1 \leq i \leq k$, $\langle \vec{v}_1, \dots, \vec{v}_i \rangle = \langle \vec{w}_1, \dots, \vec{w}_i \rangle$. If V is finite-dim, we can use the G-S process to transform any basis into an orthonormal basis.

- For $k=1$ take $\vec{w}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|}$ (normalise the length)
- Then given $\vec{w}_1, \dots, \vec{w}_k$; $\vec{w}_{k+1} = \frac{\vec{v}_{k+1} - \sum_{i=1}^k (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i}{\|\vec{v}_{k+1} - \sum_{i=1}^k (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i\|}$

Definition: Adjoints and Self-Adjointness

Let V be an IPS. Two endo.s $T, S : V \rightarrow V$ are **adjoint** if for all $\vec{f}, \vec{g} \in V$, $\langle T\vec{v}, \vec{w} \rangle = \langle \vec{v}, S\vec{w} \rangle$. In this case $S = T^*$ and S is the **adjoint** of T .

- If V a finite-dim IPS and $T : V \rightarrow V$ an endo; T^* exists and is unique.
- An endo. of an IPS $T : V \rightarrow V$ is **self-adjoint** if $T = T^*$.
- Let $T : V \rightarrow V$ be a self-adjoint linmap on an IPS V . Then:
 - Every eigenvalue of T is real
 - If λ, μ are distinct eigenvals of T with eigenvecs $\vec{v}, \vec{w}$, then $(\vec{v}, \vec{w}) = 0$
 - T has an eigenvalue.

Major Theorem: The Spectral Theorem

Let V be a finite-dim IPS and $T : V \rightarrow V$ be a self-adjoint linmap. Then V has an orthonormal basis consisting of eigenvectors of T .

- Let $A \in \text{Mat}(n; \mathbb{R})$ be symmetric. Then $\exists$ orthogonal $P \in \text{Mat}(n; \mathbb{R})$ such that $P^T A P = P^{-1} A P = \text{diag}(\lambda_1, \dots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to multiplicity.
- Let $A \in \text{Mat}(n; \mathbb{C})$ be hermitian. Then $\exists$ unitary $P \in \text{Mat}(n; \mathbb{C})$ s.t. $\bar{P}^T A P = P^{-1} A P = \text{diag}(\lambda_1, \dots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to multiplicity.

Jordan Normal Form

Definition: Jordan Blocks

For $r \in \mathbb{N}$: The **nilpotent Jordan block** of size r is the $(r \times r)$ matrix $J(r)$ defined by the rule $J(r)_{ij} = 1$ for $j = i+1$ and $= 0$ otherwise. The **Jordan block** of size r and **eigval** λ , for $\lambda \in F$, is the $(r \times r)$ matrix $J(r, \lambda)$ defined by the rule $J(r, \lambda) = \lambda I_r + J(r)$ with $DN = ND$.

Major Theorem: Jordan Normal Form

Let F be an algebraically closed field, V be a finite-dim VS, and $\phi : V \rightarrow V$ an endo. with characteristic polynomial $\chi_\phi(x) = (x - \lambda_1)^{a_1} (x - \lambda_2)^{a_2} \dots (x - \lambda_s)^{a_s} \in F[x]$, $a_i \geq 1, \sum_{i=1}^s a_i = n$ for distinct $\lambda_1, \dots, \lambda_s \in F$. Then $\exists$ an ordered basis $\mathcal{B}$ of V s.t. the matrix of ϕ w.r.t. $\mathcal{B}$ is **diag** with Jordan blocks on the diag. That is, $[f]_{\mathcal{B}} = \text{diag}(J(r_{11}, \lambda_1), \dots, J(r_{1m_1}, \lambda_1), J(r_{21}, \lambda_2), \dots, J(r_{sm_s}, \lambda_s))$ with $r_{11}, \dots, r_{1m_1}, r_{21}, \dots, r_{sm_s} \geq 1$ s.t. $a_i = r_{i1} + r_{i2} + \dots + r_{im_i}$, $(1 \leq i \leq s)$.

Theorem: Bézout's Identity

If $P_j(x) = \prod_{i=1, i \neq j}^s (x - \lambda_i)^{a_i}$, $\exists Q_j \in F[x]$ s.t. $\sum_{j=1}^s P_j(x) Q_j(x) = 1$.

Definition: Generalised Eigenspace

The **generalised eigenspace** of ϕ with eigval λ_i is $E^{\text{gen}}(\lambda_i, \phi) = \{\vec{v} \in V : (\phi - \lambda_i \text{id}_V)^{a_i}(\vec{v}) = \vec{0}\} \subseteq V$. For the **eigenspace** $E(\lambda_i, \phi) \subseteq E^{\text{gen}}(\lambda_i, \phi)$, remove the a_i term. $\dim(E^{\text{gen}}(\lambda_i, \phi))$ and $\dim(E(\lambda_i, \phi))$ are the **algebraic** and **geometric multiplicities** of ϕ with eigval λ_i

Remark: The Proof of Jordan Normal Form

This is a rough overview. If you need specifics you were cooked anyway. Let F be an algebraically closed field (can not, just need to be able to split χ_ϕ), V a finite-dim VS, and $\phi : V \rightarrow V$ an endo. s.t. $\chi_\phi(x) = (x - \lambda_1)^{a_1} \dots (x - \lambda_s)^{a_s} \in F[x]$, $a_i \geq 1, \sum_{i=1}^s a_i = n$ for distinct $\lambda_1, \dots, \lambda_s \in F$.

- Decompose V into a direct sum $V = \bigoplus_{i=1}^s V_i$, where for each i :
 - $V_i = \ker((\phi - \lambda_i \text{id}_V)^{a_i} : V \rightarrow V) \subseteq V$
 - $\phi(V_i) \subseteq V_i$
 - $\phi|_{V_i} : V_i \rightarrow V_i$ is zero on V_i for large enough m_i
- Now focus on individual V_i . These have the advantage that $\psi := \phi - \lambda_i \text{id}_{V_i}$ is a nilpotent linmap on V_i . We study a finite-dim VS W with nilpotent endo. $\psi : W \rightarrow W$, and show there is an ordered basis of W $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \dots, \vec{v}_{j_2}, \vec{v}_{j_2+1}, \vec{v}_{j_2+2}, \vec{v}_{j_2+3}, \dots\}$ s.t. the matrix of ψ w.r.t. this basis is block-diag with nilpotent J -blocks along the diag.

Move to an example for the remainder of the outline. The picture to the right describes an example with $\dim W = 16$ because each box represents a basis vector. ψ moves left to right, vanishing when it reaches the edge...
In this example the matrix has form $\text{diag}(J(5), J(5), J(3), J(2), J(1))$, and W has ordered basis (partitioned to match blocks) $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \mathcal{B}_3 \cup \mathcal{B}_4 \cup \mathcal{B}_5$

$$= \begin{pmatrix} \vec{v}_{11} & \vec{v}_{21} &$$